

statA (Dd-STATa, DICI, UniProt O00910) — Adjudication of GO:0042127 (regulation of cell population proliferation) and GO:0006952 (defense response)

Focus type: function_assignment · Hypothesis slug: proliferation-and-defense-response
Iteration: 3 of 3 (final)

Executive Judgment

Both arms of the hypothesis are **WEAKLY SUPPORTED / OVER-ANNOTATED** (paralog- and ortholog-based carry-over).

Decisive provenance (QuickGO PAINT, GO_REF:0000033, ECO:0000318/IBA): - **GO:0042127** is propagated to statA from with/from leaves = mammalian STAT3/5A/5B/1, Drosophila Stat92E, zebrafish, **and the Dictyostelium paralog STATb (dstB / DDB_G0268638 / Q70GP4)** — statA itself has **no experimental leaf**. The experimental proliferation evidence belongs to **STATb:**

P 14701681

(Zhukovskaya et al. 2004) — *"It has a subtle role in growth, so that Dd-STATb-null cells are gradually lost from the population when they are co-cultured with parental cells."* This is a textbook **paralog-based overannotation** of statA. - **GO:0006952** is propagated from with/from leaves that are **100% metazoan** — Drosophila Stat92E, mouse Stat1/2/6, rat Stat,

human STAT1 (P42224) and STAT2 (P52630) (the interferon/antiviral STATs), *C. elegans* sta-1 — with **zero Dictyostelium** leaves. Pure ortholog carry-over of the metazoan immune role.

- **GO:0042127 "regulation of cell population proliferation" — over-annotated (recommend do-not-accept as core; treat as non-core IBA lead).** In UniProt this term carries only an **IBA (Inferred from Biological Ancestor, GO_Central)** evidence code. There is **no primary Dictyostelium experimental evidence** that statA modulates the frequency/rate/extent of cell division (vegetative growth). Every experimentally-supported statA annotation concerns **developmental differentiation and morphogenesis** (culmination, stalk-cell differentiation), which the seed correctly asks to distinguish from proliferation. The IBA reflects mammalian STAT3/STAT5 growth-promoting roles, not amoebal biology.
- **GO:0006952 "defense response" — over-annotated (recommend do-not-accept as core; treat as non-core IBA lead).** Also **IBA:GO_Central only.** *Dictyostelium* innate/defense functions are mediated by **sentinel (S) cells and the TIR-domain protein TirA** (

P 17673666

), with **no reported role for statA**. The one *Dictyostelium* STAT with a stress role is the **paralog STATc** (hyperosmotic shock;

P 17517120

), which is (a) a different gene and (b) a **stress response (GO:0006950)**, not a **defense response (GO:0006952)** — the seed's "generic stress vs. defense" distinction is decisive here. The IBA reflects metazoan STAT1/STAT2 antiviral/immune roles.

Core function (what the primary evidence actually supports): statA/Dd-STATa is a **cAMP-activated STAT-family transcription factor** that acts as both a **repressor** (of stalk/ecmB genes prior to stalk-tube entry) and an **activator** (of pstA/pstAB genes such as cudA), driving **prestalk patterning, stalk-cell differentiation and culmination during sorocarp development**, with an additional early role in **chemotaxis/aggregation**

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P 10393118

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P 10821768

P 11269945

P 15470642

Evidence Matrix

Citation (PMID)	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context	Confidence/ Limit
UniProt O00910 / QuickGO (GO_REF:0000033)	Database/ annotation	Qualifies both	Evidence tier of the two seed terms	GO:0042127 and GO:0006952 are IBA-only for statA; all statA experimental (IMP/IDA/HMP) annotations are developmental/ transcriptional	Curated record	Highly probable IBA evidence
 14701681 (Zhukovskaya 2004)	Mutant phenotype (paralog STATb null)	Competing (paralog)	Which Dictyostelium STAT regulates proliferation?	STATb (dstB/ Q70GP4), not statA, has the experimental proliferation phenotype (STATb-null cells lost from population in co-culture); source of the ancestral IBA	D. discoideum growth/early dev	Highly probable to the
QuickGO with/ from for GO:0006952	Structural/ evolutionary (orthology)	Refutes/ Competes	Origin of defense-response term	with/from leaves are 100% metazoan (human STAT1/ STAT2 etc.), no Dictyostelium leaf	Cross-species PAINT	Highly orthologous carry
10393118			Does statA null affect	Dd-STATa null: delayed		Highly referred

Citation (PMID)	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context	Confidence/ Limit
	Mutant phenotype (null)	Supports core; Refutes proliferation	growth or development?	aggregation, slow chemotaxis, aberrant prestalk gene expression, fails to culminate; acts as repressor of stalk commitment. No growth/ proliferation defect reported	D. discoideum, development	
10821768	Direct assay + mutant	Supports core (TF activity)	Mechanistic TF role	Dd-STATA is a direct activator of cudA in prestalk cells and binds promoter elements	D. discoideum	High
11269945	In vitro binding + mutant	Supports core (repressor)	Repressor mechanism	Dd-STATA binds ecmB ST-promoter elements and represses stalk-tube gene transcription	D. discoideum	High
15470642	Expression profiling (in situ)	Supports core	Target-gene regulation	Identifies pstA/ pstAB genes directly	D. discoideum slug	High

Citation (PMID)	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context	Confidence/ Limit
				induced by Dd-STATa		
17673666	Discovery + mutant (TirA)	Refutes/ Competes (defense)	Who mediates Dictyostelium defense?	Sentinel cells + TIR-domain TirA provide innate immune/ detoxification defense and bacterial feeding — statA not implicated	D. discoideum slug	High the t defe mac
17517120	Mutant + microarray	Competes (defense/ stress)	Which STAT handles stress?	STATc (paralog) is the key regulator of the hyperosmotic-shock transcriptional response	D. discoideum	High stres (GO: not c
InterPro/Pfam (O00910)	Structural/ evolutionary	Qualifies	Domain architecture	STAT coiled-coil, STATa Ig, SH2 , EF-hand-12, p53-like DNA-binding fold — canonical STAT TE, no phagocytic/ immune effector domain	Sequence	High

GO Curation Implications (leads — require curator verification)

GO term	Aspect	Current evidence	Recommended action
GO:0042127 regulation of cell population proliferation	BP	IBA:GO_Central only	Do not treat as core; flag as non-core / candidate for removal for DICDI. No species-specific evidence; statA's role is differentiation, not population expansion. If retained, keep strictly as low-confidence IBA.
GO:0006952 defense response	BP	IBA:GO_Central only	Do not treat as core; flag as non-core / candidate for removal for DICDI. Defense in Dictyostelium = TirA/S-cells (P 17673666); no statA evidence.
GO:0031149 / 0031154 / 0030587 / 0045892 / 0010628 / 0097696	BP	IMP/IDA/HMP	Retain — these are the experimentally-grounded core annotations.

Curator note: the recommendation is to **downgrade/remove the two IBA seed terms as core annotations**, not to assert a NOT-qualifier without a dedicated negative experiment. Preferred, more-informative core BP terms already exist (sorocarp stalk cell differentiation; culmination in sorocarp development).

Mechanistic Scope

- **Immediate molecular function:** sequence-specific DNA-binding transcription factor (SH2-mediated cAMP/tyrosine-phosphorylation-dependent activation → nuclear translocation in prestalk tip cells).
- **Direct activity:** transcriptional repression (ecmB/stalk genes) and activation (cudA, pstA/pstAB genes).

- **Downstream/developmental outcomes (not the molecular function):** stalk-cell differentiation, culmination, fruiting-body (sorocarp) morphology, chemotaxis efficiency.
- **Neither "proliferation regulation" nor "defense response" is an immediate molecular activity nor an experimentally-observed loss-of-function phenotype** for *statA*.

Conflicts and Alternatives

- **Paralog confusion (proliferation):** The experimental "regulation of cell population proliferation" phenotype in *Dictyostelium* belongs to the **paralog STATb** (*dstB/Q70GP4*;

P 14701681

IMP), whose null cells are competitively lost in co-culture. *statA* has only IBA, propagated from the ancestral node that STATb helps define. **The proliferation function should be attributed to STATb, not *statA*.**

- **Paralog confusion (stress ≠ defense):** The stress-responsive STAT is **STATc**, not *statA* — and its role is **osmotic stress (GO:0006950)**, a different GO branch from **defense response (GO:0006952)**.
- **Ortholog carry-over (defense):** The GO:0006952 with/from set is entirely metazoan (human **STAT1/STAT2**, *Drosophila* Stat92E, mouse Stat1/2/6, *C. elegans* sta-1) with **no Dictyostelium leaf** — the term reflects the metazoan interferon/antiviral STAT role, a classic ortholog-overannotation.
- **True defense effector is unrelated:** TirA (TIR domain) and sentinel cells

(P 17673666)

provide *Dictyostelium* defense independently of STAT signaling.

Knowledge Gaps

1. **Vegetative growth rate of *statA*-null** — Not directly quantified in the retrieved literature. Matters because a documented proliferation phenotype would be the only way to legitimately support GO:0042127. *Resolve:* growth-curve / doubling-time assay of *statA*-null vs. parental in axenic and bacterial-lawn growth.

2. **statA in bacterial killing / S-cell function** — Untested. Matters for GO:0006952. *Resolve:* bacterial-clearance and S-cell assays in statA-null.
3. **GO_Central PAINT node — RESOLVED this run.** Both seed terms trace to PAINT reference GO_REF:0000033 at PANTHER node PTN000927860; GO:0042127 with/from includes paralog STATb (Q70GP4, experimental IMP) plus mammalian STAT3/5, and GO:0006952 with/from is 100% metazoan (human STAT1 P42224, STAT2 P52630). Remaining uncertainty: whether curators intend a taxon constraint on these ancestral annotations for the amoebozoan branch.

Discriminating Tests

- **Proliferation:** compare doubling time and saturation density of statA-null vs WT (axenic HL5 and on *Klebsiella* lawns). Prediction under the null hypothesis: no difference → refutes GO:0042127.
- **Defense:** intracellular bacterial-killing assay and sentinel-cell counting in statA-null; compare with tirA-null positive control. Prediction: statA-null behaves like WT → refutes GO:0006952.
- **Paralog control:** run STATc-null in the osmotic/defense assays to confirm the phenotype tracks with STATc, not statA.

Curation Leads (require curator verification)

- **Action change:** Reclassify **GO:0042127** and **GO:0006952** for statA from accepted to **non-core / remove** (IBA-only, no DICI experimental support).
- **Candidate references to cite in the review:**

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P 14701681

— proliferation phenotype belongs to **paralog STATb**, not statA (snippet: *"It has a subtle role in growth, so that Dd-STATb-null cells are gradually lost from the population when they are co-cultured with parental cells"*). Use to justify removing/de-coring GO:0042127 on statA.

- P 17673666**

— Dictyostelium defense = TirA/sentinel cells (snippet: "*A Toll/interleukin-1 receptor (TIR) domain protein, TirA, was also required for some S cell functions and for vegetative amoebae to feed on live bacteria.*").
- P 17517120**

— stress response belongs to paralog STATc (snippet: "*STATc ... is a key regulator of the transcriptional response to hyperosmotic shock.*").
- P 10393118**

— statA core developmental/repressor + chemotaxis role.

 - **Preferred core BP terms (retain):** GO:0031149 sorocarp stalk cell differentiation; GO:0031154 culmination involved in sorocarp development; GO:0045892 negative regulation of DNA-templated transcription.
 - **Suggested curator questions:** Is there any unpublished/dictyBase growth or immunity phenotype for statA? Was the STAT-node IBA reviewed for taxon restriction?

Provenance (computed artifacts)

- `statA_GO_evidence_tiers.csv` — GO evidence-tier table from UniProt O00910 (2 IBA-only seed terms isolated vs. experimental core).
 - `statA_IBA_provenance.csv` — QuickGO PAINT with/from leaves for both seed terms, showing paralog STATb + metazoan STAT1/2/3/5 as the propagation sources.
 - `statA_experimental_annotations.csv` — statA's IMP/IDA/HMP core annotations with PMIDs.
 - Domain architecture from InterPro/Pfam (STAT coiled-coil, STATa-Ig, SH2, EF-hand-12, p53-like DNA-binding fold).
 - Key computed facts: statA GO:0042127 & GO:0006952 = IBA (GO_REF:0000033) only; paralog STATb (Q70GP4) carries experimental IMP GO:0042127
- (**P 14701681**);
- GO:0006952 with/from = 100% metazoan (human STAT1 P42224, STAT2 P52630).

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